Slide 1: Solution proposal

“This slide introduces our proposed Intelligent Scan and Plan approach. The aim is to connect part identification, surface measurement, inspection and targeted repair through four stages.

First, we use a camera to identify the panel type, locate it within the workspace and select the corresponding CAD model. This establishes which part we are working with and where it is positioned.

Second, we generate an initial scanning trajectory. A detailed surface scan then captures the actual geometry, allowing us to reconstruct or update the CAD surface used for planning.

Third, we adapt the inspection trajectory to both the measured part and the requirements of the non-destructive testing sensor. This supports the inspection objective of detecting scratches and measuring their depth. Selecting the NDT sensor itself is outside the scope shown here.

Finally, the detected defect locations feed into mission planning for targeted repair. The robot would visit those locations, with trajectories adapted to operations such as sanding or polishing.

The central idea is to use measured information at each stage to guide the next. This creates a foundation for less manual programming and reusable planning across different panels.

The following slides explain the automation capabilities that support this approach.”

Slide 2: Advantages for Automation

“This demonstrator brings together vision, motion planning and simulation to make robotic inspection easier to adapt and reuse.

We are targeting three main capabilities: locating the actual part, generating its inspection trajectory automatically, and validating the behaviour in a virtual environment.

For motion generation, we are testing and integrating the stack with IRT Jules Verne, with reuse in other projects in mind. In parallel, we are evaluating suitable vision sensors for shiny and matt aluminium surfaces.

The wider value is the reusable capability we develop: path-generation software, perception functions, communication interfaces and simulation infrastructure. Benchmarking will help us identify where new AI approaches add value.

This work brings together expertise in vision, motion planning, AI, software integration and robotics simulation.

Let me start with how we capture and locate the part.”

Slide 3: Computer Vision

“The first step is to understand the actual geometry and placement of the part.

We scan the component using one or more 3D sensors and build a high-resolution point cloud. This provides the geometric information needed for referencing and path planning.

We then estimate the part’s position and orientation in six degrees of freedom, so the planning system knows where the component is within the robot’s workspace.

Sensor selection is important because shiny and matt aluminium surfaces present different measurement challenges. Alongside 3D scanning, the slide also shows sensing approaches relevant to surface inspection.

The key output is measured geometry and part placement that the planning system can use.

Once we have this information, the next step is to generate the robot trajectory.”

Slide 4: Automated Trajectory Generation

“Here, the objective is to generate the inspection trajectory automatically from the measured part.

The trajectory adapts to the component’s position and orientation, as well as geometric variations between parts. The goal is autonomous generation, without manually programming the scan points for every new panel.

The capability we are developing also addresses three further needs: on-the-fly compensation for part deformation, trajectory tuning to suit the sensor’s capabilities, and path optimization for the manufacturing process.

Reusability is central to this approach. We want to carry the same planning capability into different inspection applications, adapting the configuration and interfaces to each application.

The practical value is less manual programming for new parts and less repeated engineering across projects.

The next slide shows how simulation supports testing and validation of this capability.”

Slide 5: Digital Twin & Simulation

“The digital twin provides a virtual environment for testing sensing, robot behaviour and planned trajectories before physical execution.

It brings together realistic sensor simulation, robot-controller behaviour, contact modelling and collision checks. Synthetic data generation also supports AI training and evaluation.

The testing scope shown here includes more than one hundred parts, different part orientations and varied scratch distributions. Parallel scenario processing allows us to explore a wider range of conditions efficiently.

We can use this environment to evaluate trained AI models and check robot trajectories. A further objective is to integrate simulation into the production workflow for trajectory validation.

The expected value is earlier identification of problems and fewer physical trial runs. It also gives us an environment for investigating process improvements.

These capabilities create opportunities for collaboration across the three areas shown next.”

Slide 6: Future Collaboration & Synergy

“We see opportunities for collaboration across motion planning, vision and digital twin development.

Motion planning contributes automatic scan-path generation, adaptation to panel geometry and placement, and reuse across inspection applications.

Vision contributes sensor selection, panel identification and referencing, together with scratch detection and defect-depth measurement.

AI and digital twin development contribute realistic simulation, synthetic data and trajectory validation.

These contributions connect directly: vision provides the measured part information, planning generates the motion, and simulation helps validate the proposed behaviour.

By sharing these capabilities across projects, we can reduce repeated engineering and make new inspection applications easier to introduce.

The main value is less manual programming for new panels, supported by reusable software and validation capabilities.”